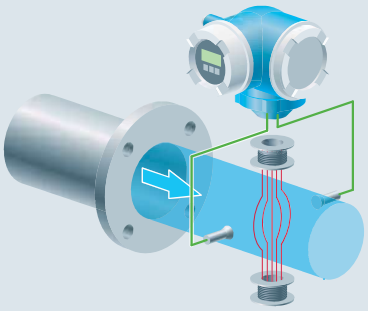
Electromagnetic flowmeters

Proline Promag – Universally applicable in all industries and in pipes from 2 millimeters to 3 meters. Since 1977, Endress+Hauser has delivered over 3 million devices.



Measuring principle

Faraday's law of induction states that a metal rod moving in a magnetic field induces electrical voltage. This dynamo principle also governs the way electromagnetic flowmeters work.
As soon as the electrically charged particles in a fluid cross the artificial magnetic field generated by two field coils, an electric voltage is induced. This voltage, tapped by two measuring electrodes, is directly proportional to the velocity of flow and thus to the flow volume. The magnetic field is generated by a pulsed direct current with alternating polarity. This ensures a stable zero point and makes the flow measurement insensitive to multiphase or inhomogeneous liquids, as well as low conductivity.

Advantages at a glance

- The measuring principle is virtually independent of pressure, density, temperature and viscosity
- Even fluids with entrained solids can be metered, e.g. ore slurry or cellulose pulp
- Wide range of nominal diameters: DN 2 to 3000 (1/12 to 120")
- Free pipe cross-section: CIP/SIP cleanable, piggable
- No moving parts
- Minimum outlay for maintenance and upkeep
- No pressure losses
- Very high turndown up to 1000:1
- High degree of measuring reproducibility and long-term stability

Measuring principle movie:
www.eh.digital/3E92q6K



The popularity of magmeters across innumerable sectors of industry continues to be unabated, further proof of the worldwide success that this measuring principle has been enjoying for more than 60 years. Magmeters can be used to measure all electrically conductive liquids above 5 µS/cm with or without solids, e.g. water, wastewater, sludge, slurries, pastes, acids, alkalis, juices or fruit mashes.

The rule of thumb for magmeters is: anything that can be pumped can also be measured – a highly valued trait in measurement technology. Typical tasks include measuring and monitoring continuous flow rates, filling and dosing as well as applications in custody transfer.

In the industrial environment, magmeters are primarily used in water management and in the processing, life sciences and food industries. In tunnel construction and mining, robust magmeters are often the only option for measuring highly abrasive ore slurries with entrained solids, sand-water mixtures, filler materials or bulk solids with the required accuracy.

FROST & SULLIVAN
2017 BEST PRACTICES AWARD
GLOBAL MAGNETIC FLOWMETERS MARKET LEADERSHIP AWARD



Promag sensors

Promag H
Food industry

- For food industries, life sciences, chemicals and process industry
- Robust stainless steel housing (3-A, EHEDG)
- CIP/SIP cleanable
- PFA liner: –20 to +150 °C (–4 to +302 °F)
- Flexible connection concept
- DN 2 to 150 (1/12 to 6")



F **L** E X
F L **E** X

Promag S
For demanding fluids

- For inhomogeneous or abrasive fluids (ore slurry, cement, paper pulp, etc.)
- Industry-optimized measuring electrodes
- Linings: PTFE, PFA, polyurethane or natural rubber
- High-temperature version up to 180 °C (356 °F)
- DN 15 to 600 (½ to 24")



F L E **X**

Promag P
Chemical and process industry

- For high fluid temperatures
- With all common hazardous area approvals
- For custody transfer
- PTFE: –40 to +130 °C (–40 to +266 °F) PFA: –20 to +180 °C (–4 to +356 °F)
- DN 15 to 600 (½ to 24")



F **L** E X
F L **E** X

Promag E
Chemical and process industry

- For the economical and cost-efficient flow measurement in basic applications
- PTFE lining: –10 to +110 °C (14 to 230 °F)
- DN 15 to 600 (½ to 24")



F L **E** X

Promag W
Water & wastewater

- For hazardous areas
- Drinking water approvals
- IP68 (Type 6P) for underground or underwater applications
- For custody transfer
- Linings: Hard rubber: 0 to 80 °C (32 to 176 °F), Polyurethane: –20 to +50 °C (–4 to +122 °F) PTFE: –20 to +90 °C (–4 to +194 °F)
- DN 25 to 3000 (1 to 120"), without inlet run



F **L** E X
F L **E** X

Promag D
Water & wastewater

- Wafer device with shorter installation length and less weight
- Drinking water approvals
- Lining: Polyamide: 0 to 60 °C (32 to 140 °F)
- DN 25 to 100 (1 to 4")



F **L** E X

Magphant
Limit switch

- For cost-effective flow monitoring
- For steel or plastic pipes
- DN 15 to 2000 (½ to 80")



Another highlight with Endress+Hauser magmeters is the flexible installation independent of flow profile and without pressure loss, e.g. directly after bends.

Find out more in this animation:
www.eh.digital/3FgnTeB

Proline transmitters

Proline – the perfect transmitter for every application
Whether applied in the food, water, chemical, mining or other process industries, the new generation of Proline transmitters can be freely combined with many of our sensors, all tried and tested for decades – while maintaining full backwards compatibility. This means a high level of added value and maximum flexibility matching your industry and application demands.

Detailed description of all Proline transmitters ▸ page 9 to 11

Promass sensors	Proline transmitters							
	10	100	200	300	400	500		800*
								
Promag H	✔	✔	✔	✔	–	✔	✔	–
Promag P	✔	✔	✔	✔	–	✔	✔	–
Promag E	–	✔	–	–	–	–	–	–
Promag W	✔	–	–	✔	✔	✔	✔	✔
Promag D	✔	–	–	–	–	–	–	–

* battery-powered

reddot design award
winner 2018



Picomag
Flow metering in a pocket-sized format

- Reliable measuring and monitoring of industrial water, cooling water or warm water
- Simultaneous measurement of flow, temperature and conductivity
- Flexible integration into all fieldbus systems using IO-Link
- Intuitive commissioning and operation using Bluetooth and the SmartBlue app
- Cost-effective measuring device for the use where space is at a minimum

